NEWTON'S LAWS OF MOTION

Newton's 1st law:

Everybody continues its state of rest of ~~n~~ or uniform motion along straight line until it ~~experiences~~ acted on by an unbalanced external force or torque.

It is also known as law of ~~motion~~ inertia. (2 types)

Inertia: Tendancy of an object to continue its state of rest or uniform motion along straight line & linear motion.

→ It is of 3 types:

1. Inertia of motion

It is defined as the property by ~~virtue of~~ which a body in motion continues to be in motion unless acted upon by an external force.

eg: 1: peddling the bicycle it move to a distance ~~wh~~ without peddle.

2: A car continues to move forward even its engine has stopped.

3: ~~Athle~~ In a sport field an athlete often runs before taking a long jump.

4: The fan continue its motion even after it is switched off.

5: passenger in a running bus tends to lean forward when the bus stops.

2. Inertia of rest

It is defined as the property by virtue of which a body continues at rest unless acted by some external force to change that condition.

eg 1: fall of coin kept on a cardboard into the glass.

2: fall of rider backwards

3: fall of dust particle from a cloth.

4: when you are standing in a stationary bus, it suddenly starts, you fall backward.

5: when step on an elevator we experience a forward force and falls backward.

3. Inertia of position (direction).

It is defined as the inability of a body or object to change its position / direction of motion by itself.

ie, external force is required to change its direction of motion.

eg 1: One's body movement to the side when a car makes a sharp turn.

2: Tightening of seatbelt in a car, when it stops quickly.

3: Men in space find it more difficult to stop moving because of a lack of gravity against them.

4: Throwing of shortput.

5: When the direction of the whirlpool is changed, it tends to continue in the same direction ~~even if the~~ & for a while before changing.

Newton's 2nd Law.

It is defined as the rate of change of momentum of an object directly proportional to the force applied on it and takes place in the direction in which the force acts.

eg: ~~1: catching of a cricket ball~~

1: A cricketer lowers his hands while catching the ball

2: A girl lower ~~her~~ hands while playing with ring

3: If same force is used to push the car and trunk, the car will have more acceleration than truck because the car has less mass.

4: It is easy to push an empty shopping cart than a full one, because it requires more force to push the filled shopping cart.

5. when an object falls in a free fall onto the ground, it accelerates because the force of gravity of earth pulls it.

3. Newtons 3rd law.

It is defined as for every action, there is an equal and opposite reaction.

eg: 1: rocket repulsion

2: Recoiling of gun.

3: Jumping to land from boat

4: walking : action:- apply force on street.

Reaction:- force move you forward

5: when air rushes out of a balloon, the opposite reaction is that the balloon flies up.

Moment of Inertia.

It is the property of an object that resist rotatory motion and changes in rotatory motion.

The law of inertia indentifies the condition at which the an object is in equillibrium. equillibrium can be static (@ rest) or dynamic (@ motion).

If the force acting on an object is balanced, the object can be motionless it is called static equillibrium.

when velocity of an object is constant but not zero, the object is moving and is in dynamic equillibrium.

Dynamic motion can be linear - (translatory) around an axis -[Rotatory motion] or in combination of both (curvilinear motion)

Thus newton's 1st law of motion helps us in determining whether the object is in static or dynamic equillibrium.

The newton's law of an inertia thus also be called law of equillibrium. It is restated as, for an object to be at equilibrium the sum of forces applied to it must be zero.

$\Sigma F = 0$

$\Sigma \tau = 0$

Newton's 2nd law.

* It is also known as newton's law of accelleration.
* The law also can be Stated as linear accelleration or angular

acceleration of an object is $\propto$ to the net unbalanced force or (τ) acting on it. further, acceleration is inversely proportional to the mass.

$$a = \frac{F_{unbalanced}}{\text{mass } M}$$

$$\alpha = \frac{\tau_{unbalanced}}{I_{\text{moment of inertia}}}$$

→ An object acted on by a net unbalanced force or τ must be accelerating, An acceleration of object will be in the direction of net unbalanced force or τ.

→ A net unbalanced force will produce translatory motion.

→ A net unbalanced τ will produce rotatory motion.

→ An the combination of both produce curvilinear motion.

Newton's 3rd law.

→ Also known as law of reaction. It is stated as when an object applies force on a 2nd object, the 2nd object must simultaneously apply a force equal in magnitude and opposite in direction to the first object.

→ The 2 forces that are applied to the 2 contacting objects are an interaction pair and can be called as action-reaction forces or simply reaction forces.

→ Reaction forces are always in same line and applied to different but contacting objects.

→ Direction of 2 forces are opposite to each other. because the direction of reacting forces is always opposite to each other

DESCRIPTION OF THE MOTION

→ There are 5 kinematic variables that fully describes the motion.

i) The type of displacement (motion)
ii) The location in space
iii) The direction of displacement
iv) the magnitude of displacement
v) The rate of change in displacement.

1. Type of displacement.

→ 3 types i) Translatory - linear
ii) Rotatory - angular
iii) General motion.

i) Translatory

* Here movement occurs in a straight line. In true translatory motion, each point of a segment moves through same distance and same direction.

* In Human motion isolated translatory movements are rare.

* eg: Anterior Drawer test for anterior cruciate ligament integrity.

ii) Rotatory or rotary

* Here movements occur around a fixed axis (called centre of rotation) in a curved path.

* In true rotatory motion each point on the segment moves through same angle, at the same time, at a constant distance from the centre of rotation.

* eg: True rotatory motion can occur only if segment is prevented from translating and is forced to rotate around a fixed axis. This rarely happens in human motion.

iii) General motion

* combination of rotatory and translatory movement are seen in human body. Such movements are called general movement.
This motion can also be called, curvilinear motion (plane or planar motion).

* Curvilinear motion is a combination of translation and reaction in 2D. When this type of movement occur, The axis is not fixed but it shifts as the object moves.

* The axis around which the segment appears into move in any parts of its motion is reffered to as instantaneous axis of rotation or instantaneous centre of rotation.

* 3D motion. It is a general motion in which in the segment moves across all 3D or planes. The axis around which 3D motion occur is called, Helical axis of motion or Screw axis of motion.

2. Location of Displacement

Rotatory or translatory motion of body Segment is can be commonly located in shaped by using 3D cartesian cordinate system, using x, y and z axis. It is used assuming that body is in an anatomic position and centre of mass of the body as the origin of x, y and z axis.

→ X-axis

If body is at anatomic position, its axis runs from side to side of body, and is labelled as coronal axis.

→ Y-axis

It runs from up to down of the body and is labelled as vertical axis.

→ Z axis

It runs from front to back or antero-posterior, and is labelled as A-P axis.

* Motion of a segment can occur either around an axis or along an axis.

* A segment can either rotate or translate around each of the 3-axis, Thus resulting in 6 potential option for that segment.

* The option for movement of the segments are reffered ~~as~~ to as degrees of freedom.

* A completely unconstrained segment always has 6 degrees of freedom. But segments of the body are not unconstrained. Hence, have less than 6 degrees of freedom.

* Rotation of a body segment occurs ⊥ to the axis and ‖lel to the plane.

* There are 3 planes around which body movements occur. They are
sagittal,
transverse and
frontal plane.

* Sagittal plane motions are said to be as ~~side to side motion~~.
front to back motion of a segment.
eg: flexion - extension of shoulder joint.

* Transverse plane motions are said to be as parallel to the ground.
eg: medial - lateral rotation of shoulder joint.

* Frontal plane motion are said to be as side to side motion.
eg: abduction - adduction of shoulder joint.

Motion of a segment always occur parallel to plane, ⊥ to corresponding axis.

* sagittal plane ⊥ coronal axis
* frontal plane ⊥ A-P axis
* transverse ⊥ vertical axis
* eg: shoulder flexion and extension
plane - sagittal axis coronal

~~→ shoulder abduction - adduction~~

BASIC BIOMECHANICS

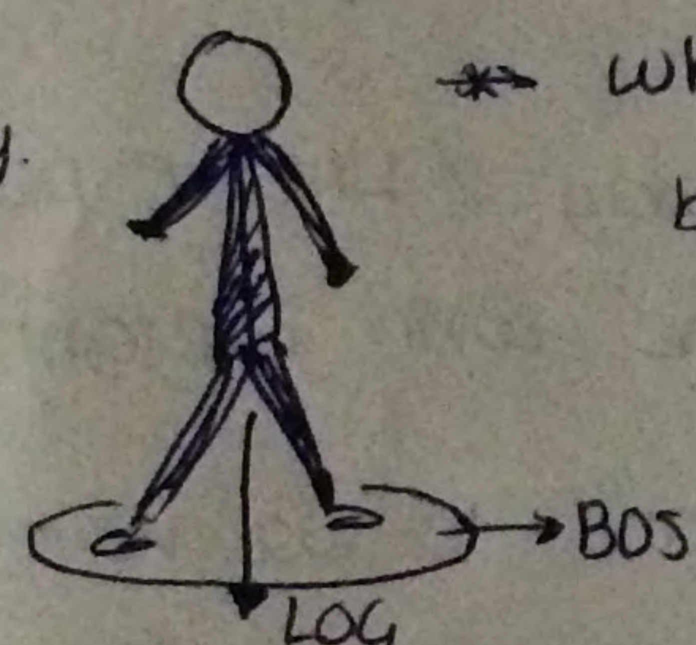
1. Gravity : invisible force, that acts on every objects by the earth.
   - it is the force that pulls the body to the earth ~~most~~
   - It is most consistent, influential ~~and~~ invisible. and natural phenomenon.
2. COM/COG : assumes a point, entire mass of an object is assumed to be concentrated. (hypothetical point)
   - If object is symmetrical, COM is concentrated @ centre.
   - If object is assymetrical, COM will be at the heavier side.
   - If an hollow object, acts outside the body.
   - Balance object even with a finger.
3. LOG : every force has a point of application (COM), direction (always downward & LOG) and magnitude.
   : It is gravitational force vector of a body / Line of pull of gravity.
   : can be compared as a plump line with weight suspended.
4. COM human being : located approximately ant to S_2 vertebrae in anatomical position.
   : COM can vary acc to variation in position of the body.
   : In newborn : COM located above umbilicus
   6 month : T_6
   2 yrs : at level of umbilicus
   5 yrs : below umbilicus
   adult : ant to S_2 vertebrae.

Base of support and stability

BOS - Base of support.

* COM - hypothetical point in which entire mass of body is concentrated, approximately lies ant to S2 vertebrae.
* LOG - vertical line / gravitational force vector.
* BOS - defined as the entire area below the person or object, in to which he or the object contacts with the ground or the supporting surface.
  - ~~In unilateral stance~~ - single limb support.
  - In bilateral stance with wider legs, there is a greater BOS, thus stability also ↑ses.
  - for old people, standing or walking with wider legs, helps to increase BOS and thereby stability of the body.

chance to fall.
↑ instability.



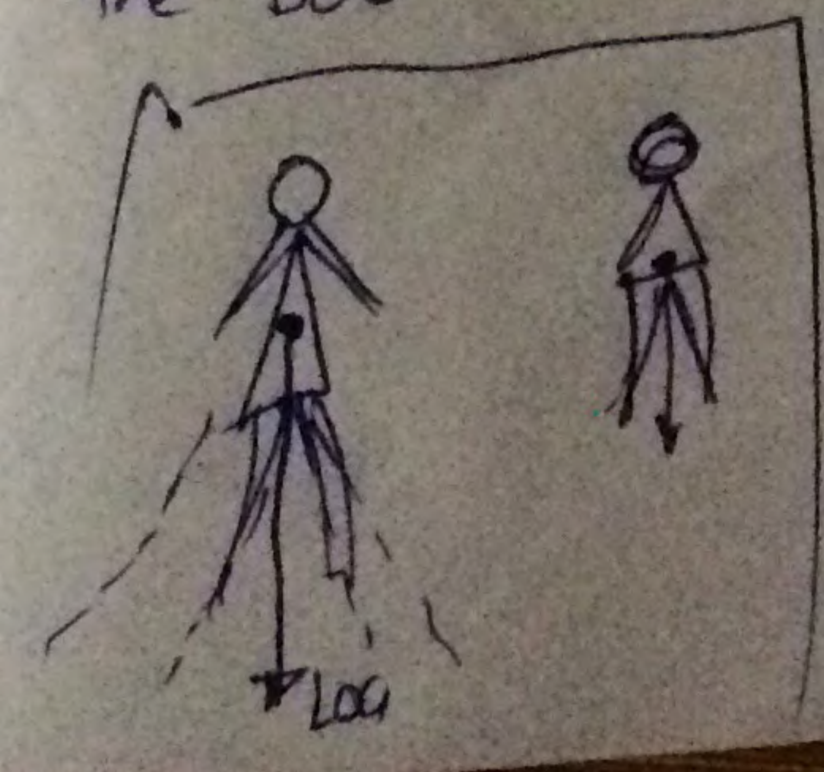
* whenever the LOG lies inside the BOS - the body will be stable.
  * In leaned position, the LOG shifts away from BOS, this results in fall of the body.

this can be prevented by altering the BOS. so that LOG still resides in the BOS.

location of COM and stability

* when COM of a body is lower to the body, the stability is more.
  eg: sitting on a chair, because LOG is more close to ground and cannot fluctuate to a greater extend.

(when COM is higher, LOG can ~~deviates~~ fluctuate in a greater extends from the BOS)



BOS can be increased by (when LOG is altered)

* hand act as BOS along with foot
* ↑sed acceleration.

Arthrokinematics and osteokinematics

* **Ostiokinematics** – there are the movements of the bone. ~~is known~~ as (as a lever) / Bony lever as a whole.
  - they are normal physiological movements of the body.
  - these are seened with naked eyes. (visible)
  - In general, mostly one of the bony segment will be fixed and the other will be mobile.

* The osteokinematic motions can be represented by few factors such as;

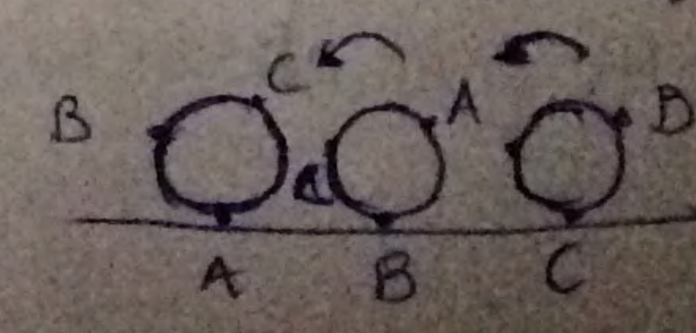
  - axis of movement – x - axis
  - plane – sagittal plane
  - direction of movement. – flexion - anteriorly.

  eg: shoulder flexion

* **Arthrokinematics** : these are the movements of articular surfaces within the joint.
  - It is normally not visible.
  - It consists of a moving and a stationary segment usually it occurs as the movement of mobile segment on the stationary segment

* The arthrokinematic motion is divided into 3 as,
  - rolling movement
  - gliding movement
  - spin.

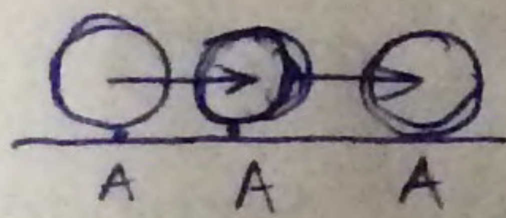
- **Roll** : ~~the point of contact~~ It is a movement at which, a ~~pt~~ every time a new point comes in contact with the articulating surface.
- Here the movement and the opposite (stationary) segment always moves.
- occurs only during pure rolling.



eg : During knee flexion. (every time new point of femur comes in contact with the tibia).

- In human body mostly rolling occurs along with gliding.

- Gliding : (sudden break). The point of contact will be always same throughout the motion.



eg : In knee flexion, posterior roll occurs along with anterior glide.

: Gliding is also known as sliding.

→ In normal human body - roll and glide occurs simultaneously.

- Spinning - the motion occurs around a fixed axis.
  - eg : motion of radius during supination and pronation.
  - In human body no pure spin occurs. It is always accompanied by roll and glide.

Convex - concave rule

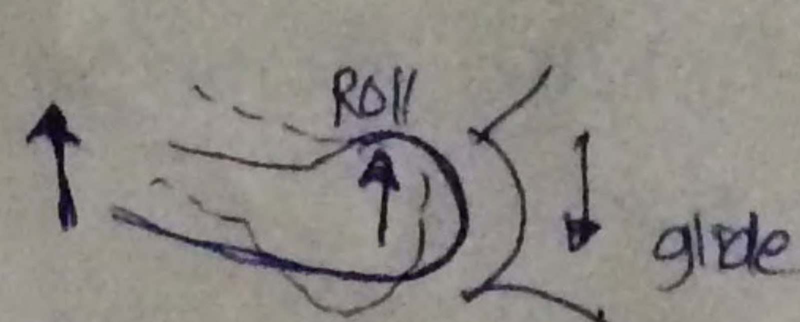
* Human movements are of 2 types — osteokinematic movement - bony lever
  — arthrokinematic - intraarticular / accessory movements.
  includes : ~~roll~~ roll / glide / spin.

[* In human body roll and glide occurs simultaneously]

* normally rolling occurs in same direction of osteokinematic motion.
* whereas glide / spin ~~eith~~ takes place in opposite direction.

→ Senario 1 : when convex part is moving on a concave,

osteokinematic - superior motion
rolling - superior
glide - inferiorly

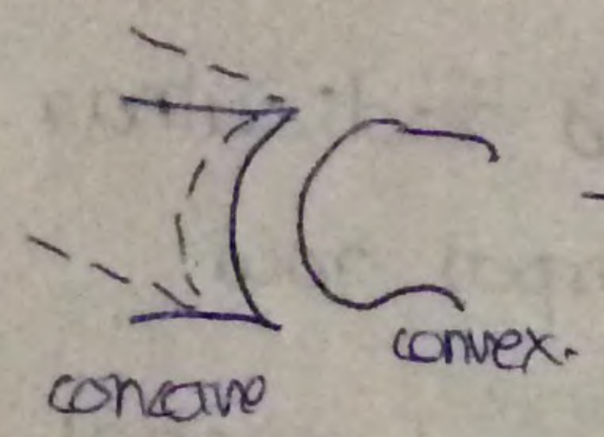


Ie, ~~the aethrokinematic~~ when a convex moving on concave osteokinematic and aethrokinematics (glide) occurs in the opposite direction.

eg : arm ~~ab~~ abduction - the convex humeral head moves on relatively fixed concave glenoid fossa
- Here osteo and atheo (gliding) occurs in opposite direction,

inorder to maximise the ROM available.

Senario -2 when concave surface moves on the convex surface, the rolling and gliding occurs in the same direction.



— roll and glide in same direction, inorder to achieve possible ROM.

or, osteokinematics and arthrokinematics occurs in the same direction.

eg: of hip bone during hip flexion (osteo), - anteriorly

Rolling - anteriorly
glide posteriorly

~~hip abd~~

eg: Tibio femoral joint

1. ~~post~~ Flexion - ~~flexing~~ flexion - posteriorly (osteo)
femur is moving. Rolling - posteriorly
glide - anteriorly.

2. flexion of tibia - glide and roll occurs in posterior direction.
(Fixed femur.

FORCES AND FORCE VECTORS

→ **Force** : defined as a push or pull on an object or when 2 objects comes in contact with each other, both of the objects exerts a force (push or pull) on each other.

* F = mass x acceleration.

(force is α to mass and acceleration)

* unit = $kg\,m/s^2$ or SI unit of force is newton / pounds (lb)

→ **weight** of a body = mass x gravity

* kg to newton 1 kg = 9.8 N

1 kg = 2.2 lb

→ **Types of forces** : 2 types → internal force - eg: ~~gravity~~ (N), ligaments, etc..
→ external force - eg: gravity, wind, water, etc.....

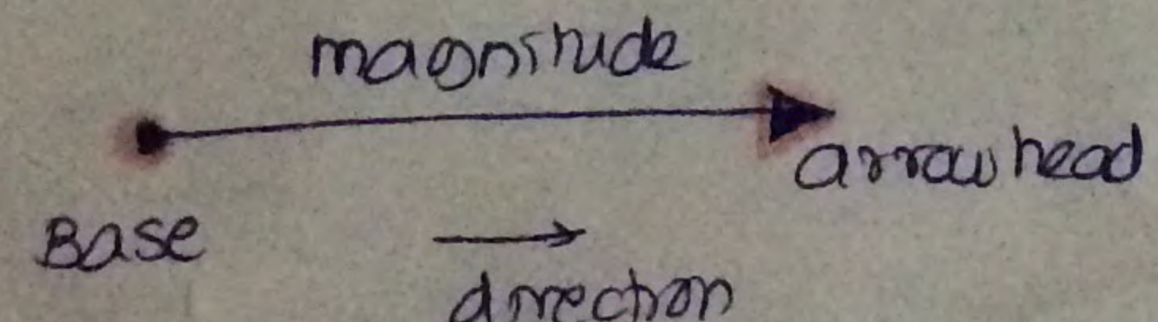
* external force can either restrict or (produce) cause the motion.
* internal forces initiates the motions and are also capable of remodifying the forces. (accelerating)

→ **force vector**

* force is a vector quantity.
* vector consists of mainly 3 parameters - 1. Base - point of application of force.

2. Arrow head and a ~~direction of~~ shaft, which shows direction of force.

3. magnitude - amount of force acted.



* Gravity on hand.

ACTION - REACTION FORCES

* action - reaction forces are based on the 3rd law of newton
* 3rd law says that for every action there is an equal and opposite reaction.

x, whenever 2 ~~forces~~ objects contact each other if object A exerts a force on 2nd object B ~~must~~ then object B must also exert a force which is equal and opposite to object A.

This forces are known as interaction forces or reaction forces or action - reaction forces.

→ characteristics

* objects should contact each other.
* whenever two objects are in contact, they must exert contact forces.
* they are in same line and opposite to each other.
* since they are acting on 2 objects they donot form part of any force system.
* They cannot be resolved.

→ contact forces

* It is a part of action - reaction forces.
* It exist if 2 of the objects are pushing each other
* If forces are ⊥ to each other, they are known as normal force.

LINEAR FORCE SYSTEM

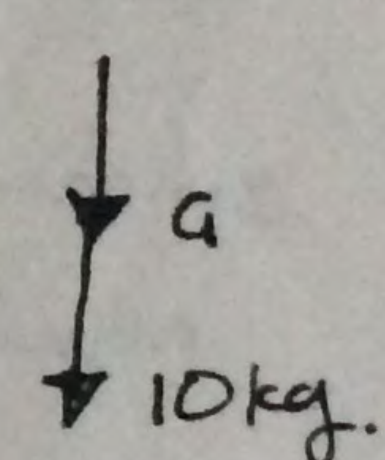
* 2 or more forces acting on a same segment are in same line that is collineae and same plane thats coplanae then the forces are said to be part of lineae system.

* ~~when the forces overlap each other.~~

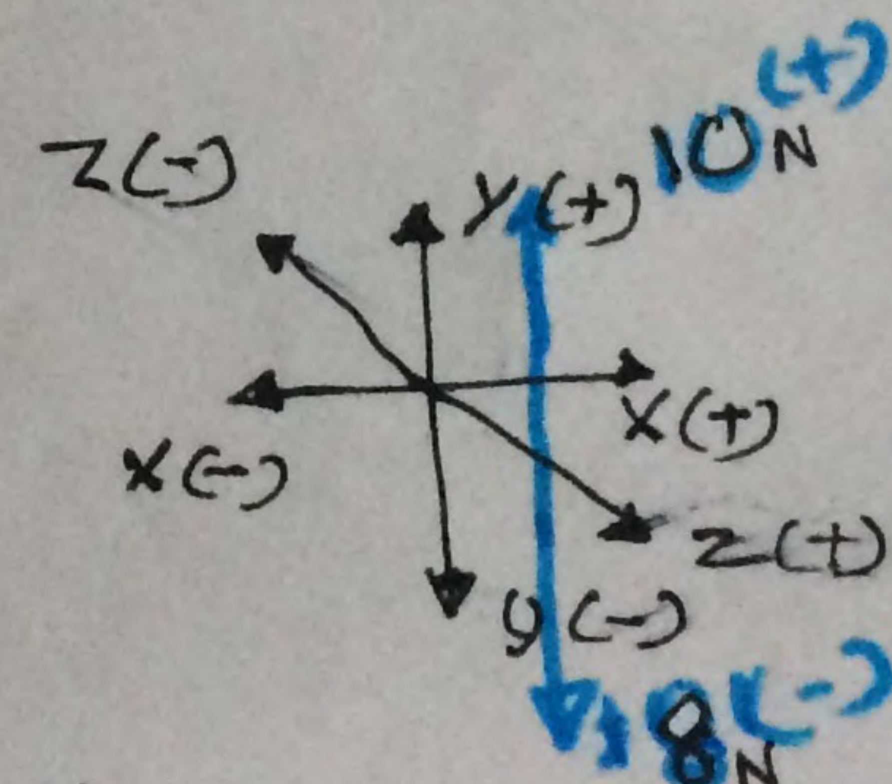
* characteeastic features are;

  - collineae
  - coplanae
  - acts on same segment
  - when extended they overlap each other.

* Further the forces in lineae force system are assigned +ve and -ve values accoeding to the direction of forces.



* The resultant ~~forces~~ of lineae force system is just the aeithmatic sum of the forces.

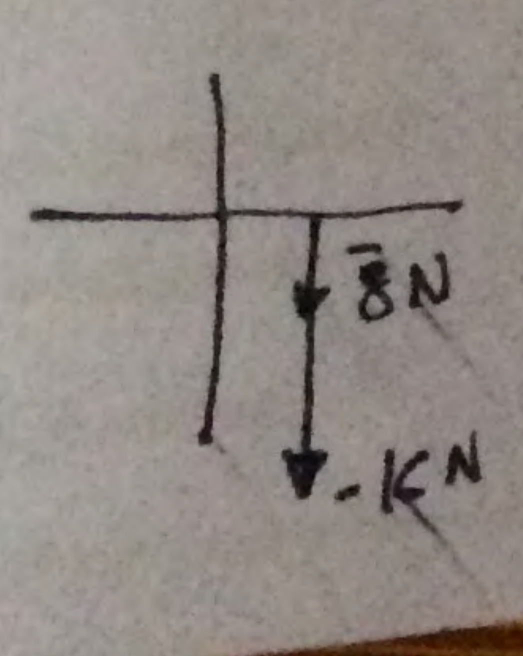


eg: resultant of

$10 - 8 = \underline{2\ N}$

eg: 2 forces acting on same segment

$-8 - 16 = \underline{24\ N}$.



eg: $-14 - 48 = \underline{\underline{-88\ N}}$

* The resultant of lineae foece is in line with the foee. and will be moee close to the highee feece.

→ Lineae force system in human body.

- tensile force system.
- joint distraction
- joint compression foeces
- etc...

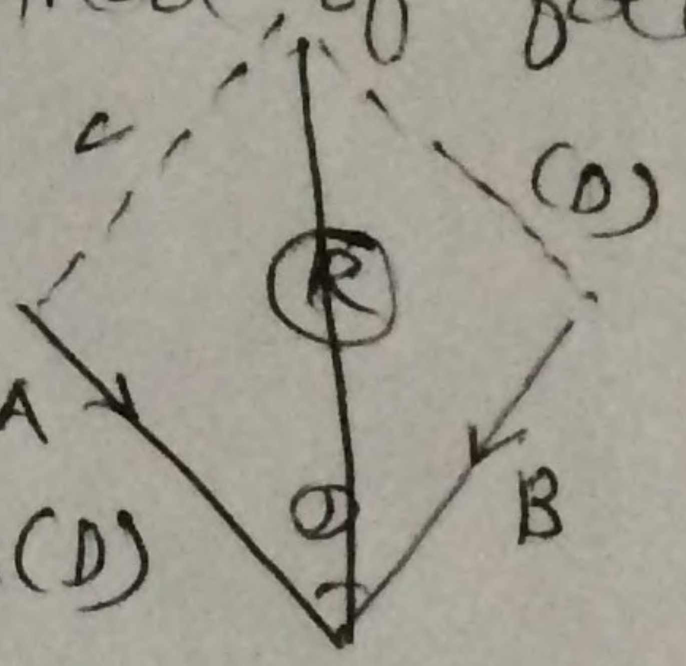
Concurrent force system.

* 2 or more forces acting on the body will forms angle with each other.
* when there is an angle exist b/w 2 or more forces acting on a same segment.
* concurrent forces lie in different plane & diff lines.

when 2 or more forces act on an ~~object~~ same segment lie at an angle to each other. such forces are known as concorrent force or concurrent force system.

→ Resultant of concurrent force system.

* finded by parallelogram method of force composition.



- draw a line parallel to force B (C)
- draw another line ∥ to force A (D)
- the diagonal of parallelogram forms the resultant.

* magnitude and resultant can be obtained through this method.

→ characteristics

- resultant will be acting in ~~b/w the forces.~~ from the point of application.

- when we extend forces in concurrent force system It intersect with each other.
- value of the resultant will be less than sum of the 2 forces.